JOM, Vol. 73, No. 6, 2021
https://doi.org/10.1007/s11837-021-04592-3
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# Using Chlorine Gas to Recover Rare Earth Metals from End-of-Life Permanent Magnets

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We describe a laboratory-scale process for recycling rare earth metals from end-of-life rare-earth/iron/boron alloy permanent magnets using a 2-h chlorine gas treatment at 673 K. This treatment does not require any special preparation of the magnets; they can be used without demagnetization, crushing or milling. Following treatment at 673 K, a clinker powder consisting of rare earth metal chlorides with minimal amounts of other metals is obtained, while hematite and iron oxychloride sublimate. Following literature sources, we suggest that the rare earth metal chlorides in the clinker may be readily reduced to metal, either by electrolysis in relatively low temperature eutectic melts or by metallothermic reduction in mixtures with alkali or alkaline earth metals.

## INTRODUCTION

Since being introduced in 1984, magnets based on a neodymium (+/− praseodymium)–iron–boron alloy have become the permanent magnets of choice (henceforth, Nd magnets) for a variety of electromechanical and electronic devices due to their superior magnetic energy product, theoretical maximum 512 kJ/m$^3$.[1] To raise its Curie temperature, the alloy may also be doped with 5–10 mass% of a third rare earth element (REE), i.e., dysprosium (Dy).[2] Given that Dy is a particularly costly rare earth metal, Pathak et al.[3] attempted to eliminate the need for Dy by doping with carefully controlled amounts of Ce and Co, producing a $(Nd_{0.8}\ Ce_{0.2})_{2.4}\ (Fe_{12}\ Co_2)B$ alloy. Currently, manufacture of Nd magnets is increasing worldwide at a rate of ~ 20% per year[4] as the need for electric generators in wind turbines and motors in hybrid/electric vehicles becomes more widespread. This has given rise to large quantities of waste REE, which may be divided into two categories: (1) manufacturing scrap and (2) end-of-life (E-o-L) industrial products. Due to the mechanical fragility of the Nd-Fe-B alloy, up to 40 mass% goes to waste scrap during magnet manufacture: pieces, shavings, powders and grinding waste.[5] Of this scrap, ~ 30 mass.% is REE.[6] Nd magnets in E-o-L devices pose an even more difficult recovery problem, since the magnets must first be removed from the hybrid/electric vehicle motor, mobile telephone, wind turbine, etc., prior to recycling treatment. Because the highly nonuniform geological distribution of RE metals cannot guarantee secure and interrupted access in all countries, development of an economical and ecologically sound recycling strategy for both types of waste is an urgent global need.

During treatment of magnet manufacturing waste or E-o-L device magnets, the major challenge is to separate RE metals from iron. There are currently two pathways for recycling valuable RE metals: pyrometallurgy and hydrometallurgy.[7–10] However, both are multi-stage processes. For E-o-L device magnets, preliminary steps include separation of the magnetic material from plastics and from metallic, but non-magnetic, components, e.g., clamps, clips, etc.; demagnetization *via* heat treatment at 573−623 K;[11] decarbonization (for removal of resins) by combustion at 973−1273 K under air or oxygen flow;[11] and deoxidization by reduction with hydrogen.[12–14] Separation of RE metals from iron can only begin after all these preliminary stages have been successfully completed. Several wet processes, e.g., acid dissolution,[15] solvent extraction and the oxalate method,[16] have been developed to

(Received October 15, 2020; accepted February 11, 2021; published online March 9, 2021)

recover RE metals. However, these wet chemical methods give poor yield from the acid dissolution and effluent treatment steps, which in turn require a multi-stage, costly process for neutralization of the acid waste. Given these difficulties, protocols for recovery of RE metals must themselves be as rapid and economical as possible, since separation of a RE magnet from the device in which it was used is currently a time-consuming process.

There are a number of reports in the literature describing pyrometallurgical extraction of RE metals from permanent magnets *via* chlorination at high temperature (600–1300 K) with $NH_4Cl$,[17] $FeCl_2$,[18] $MgCl_2$[19] or chlorine gas plus carbon, i.e., carbochlorination.[11] The use of $NH_4Cl$, $FeCl_2$ and $MgCl_2$ as chlorinators leads to the formation of neodymium chloride while the Fe/B alloy remains in solid form. The resulting mixture of neodymium chloride and Fe/B metal residue is then separated by vacuum distillation or magnetic separation.[18] The chlorination method simplifies the overall process and reduces the amount of toxic effluent requiring subsequent treatment. However, one serious disadvantage is that it is energy intensive in both the high temperature phase and during vacuum distillation.[20] The method based on carbochlorination of Nd magnets[11] demonstrated that preliminary oxidative sintering in air at 373–1273 K, with the resulting conversion of Fe and Nd to oxides ($Fe_2O_3$, $FeNdO_3$, $Nd_2O_3$), dramatically reduces the degree of sublimation of iron and boron observed during chlorination in the absence of carbon. However, when carbon was added to the chlorination process, the extent of sublimation of iron and boron chlorides was observed to increase. It is known that the Gibbs energy of the chlorination reaction of metal oxides by chlorine gas is large and positive; therefore, successful chlorination with pure chlorine is practically impossible.[21]

In the present study, our objective is to present a rapid and economical process for separating RE metals from iron via treatment of end-of-life, permanent magnets with chlorine gas at temperatures which are only moderately elevated above ambient. We have restricted our study to sintered alloy magnets; the suitability of our protocol for extruded, bonded magnets will be described in a future report. Currently, industrial magnet recycling/processing is not performed on a large scale, although with increased demand for REs, this situation is expected to change. With respect to the impact on the environment and recycling, we take as a model those chemical and metallurgical industries that produce sodium, magnesium, titanium, tantalum and niobium metals with reactions requiring large volumes of chlorine gas.[22,23] Protocols for the scrubbing of gas mixtures containing chlorine are published by the World Chlorine Council[24] with benign end products such as calcium chloride and sodium chloride. In our laboratory-scale process, most of the chlorine gas used for the reaction can be returned to the chlorination reactor following oxidation of sublimate ferric chloride.

## MATERIALS AND METHODS

### Materials

Fragments of two end-of-life, sintered REE-Fe-B alloy magnets were received from the Korea Institute of Industrial Technology. They had been removed from end-of-life hard disc drives (HDD). Although the magnets had been metal-coated to minimize corrosion, they were polished prior to being shipped to Israel. The fragments were irregularly shaped with dimensions on the order of 20–30 mm. One of the magnets, denoted sample #1, showed extensive surface corrosion (Fig. 1a); the second, sample #2, had an essentially uniform surface (Fig. 1b). In the interest of safety, chlorine gas was synthesized in-house according to the reaction:

$$Ca(ClO)_2(s) + 4HCl\,(l) = CaCl_2(s) + 2H_2O(l) + 2Cl_2(g). \quad (1)$$

$Ca(ClO)_2$ (general purpose grade) and HCl (analytical grade) were purchased from Holland Moran Ltd. (Israel).

### Methods

X-ray diffraction (XRD) of the magnet fragments was performed on a TTRAX III theta-theta diffractometer while the clinker powders and volatile materials were measured on the TTRAX instrument as well as on an Ultima III theta-theta diffractometer (both manufactured by Rigaku Corp., Japan). Phase identification and quantitative phase analysis were accomplished using Jade Pro (MDI, Cal.) software and the Inorganic Crystal Structure Database (ICSD). The Cu $K_\alpha$ X-radiation, Bragg-Brentano protocol, diffracted beam monochromator and variable slit widths constituted our standard operating procedure on both diffractometers. The elements present in the magnet fragments, clinker and sublimate powders were characterized by energy-dispersive x-ray fluorescence spectroscopy on a LEO Supra scanning electron microscope (SEM). Quantitation and material balance were provided by ICP-MS (Agilent 7700s ICP-MS System) measurements following 24-h dissolution of the magnets and clinker powders in aqua regia at room temperature with continuous stirring.

### Chlorination of Magnet Fragments

Heating of REE-Fe-B metal alloy magnets in the presence of flowing chlorine gas (80–100 ml/min) was carried out in a temperature-controlled

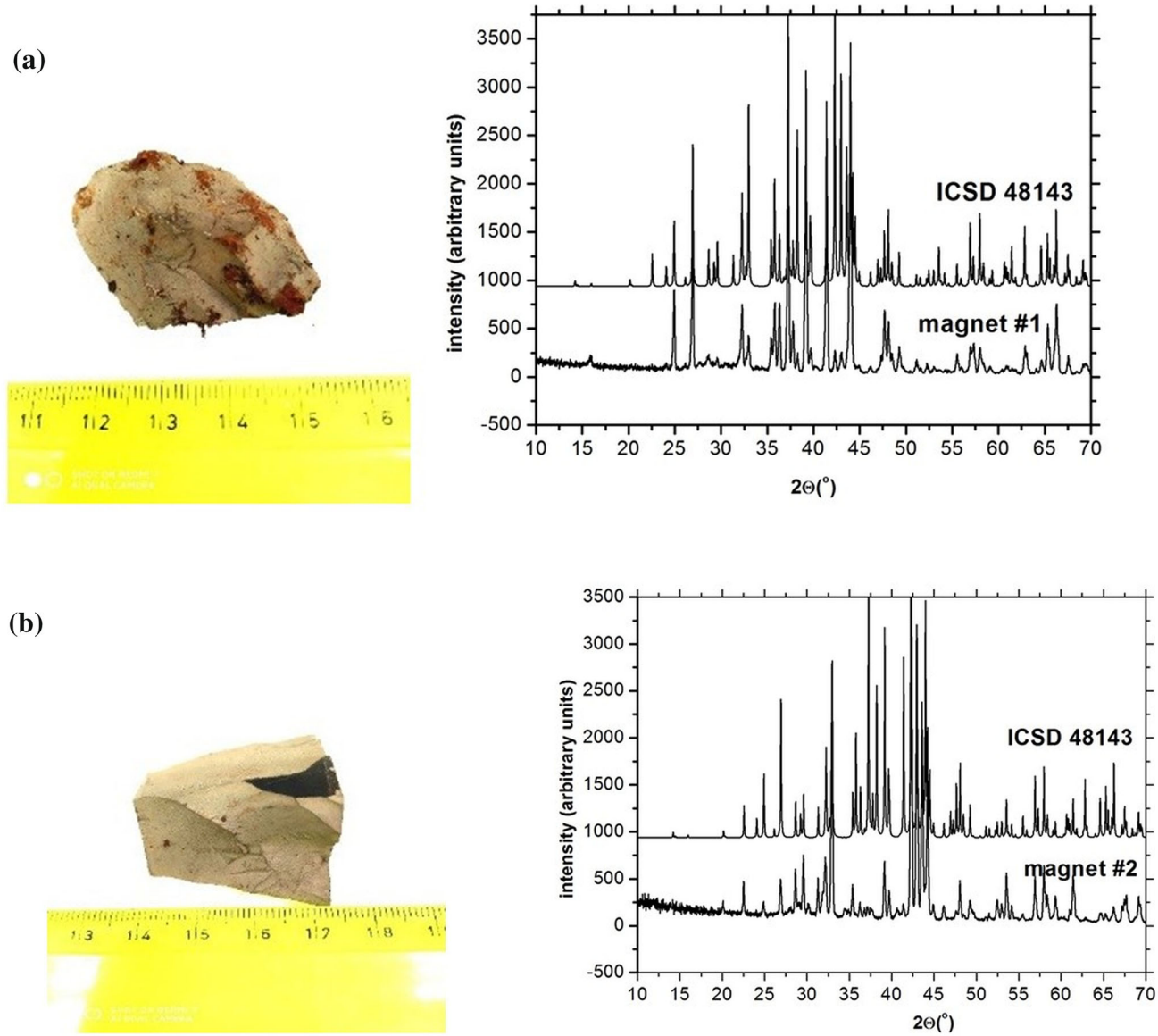



Fig. 1. Brightness-enhanced photographs and x-ray diffraction (XRD) patterns measured on the undamaged surface regions of the as-received magnet particles. (a) sample #1; (b) sample #2. The ICSD x-ray diffraction pattern for the $Nd_2Fe_{14}B$ standard alloy (#48143) is shown for comparison. Penetration depth of Cu K$\alpha$ radiation, $\sim 10\mu$m.

laboratory furnace. The laboratory setup is shown in Fig. 2. Fragments of each magnet (as received, without demagnetization, crushing or milling, but following surface polishing before shipment from Korea to Israel) were placed in the furnace in a Pyrex glass crucible (total weight approximately 25–30 g). Prior to heating, the quartz reactor with the magnet fragments in place was cleaned with a 100 ml/min flow of dry nitrogen gas for 15 min. The furnace was then heated at a uniform rate from room temperature to 673 K (during 60 min), again in the presence of 100 ml/min nitrogen flow. After the furnace had reached the designated temperature, chlorine gas was fed into the reactor for 2 h.

## RESULTS AND DISCUSSION

### Characterization of the Nd Magnet Fragments Prior to Chlorination

The as-received Nd magnet fragments were photographed (Fig. 1a and b), showing irregular shapes, 20–30 mm dimensions and prominent surface corrosion in the case of magnet #1; magnet #2 presented a more or less uniform surface. XRD profiles (measured on a relatively undamaged surface region,) were compared with the Inorganic Crystal Structure Database (ICSD) standard XRD pattern for crystalline $Nd_2Fe_{14}B$ alloy (#48143). We note that CuK$\alpha$ radiation is only able to probe a $\sim$ 10-micron layer at the magnet surface because of the

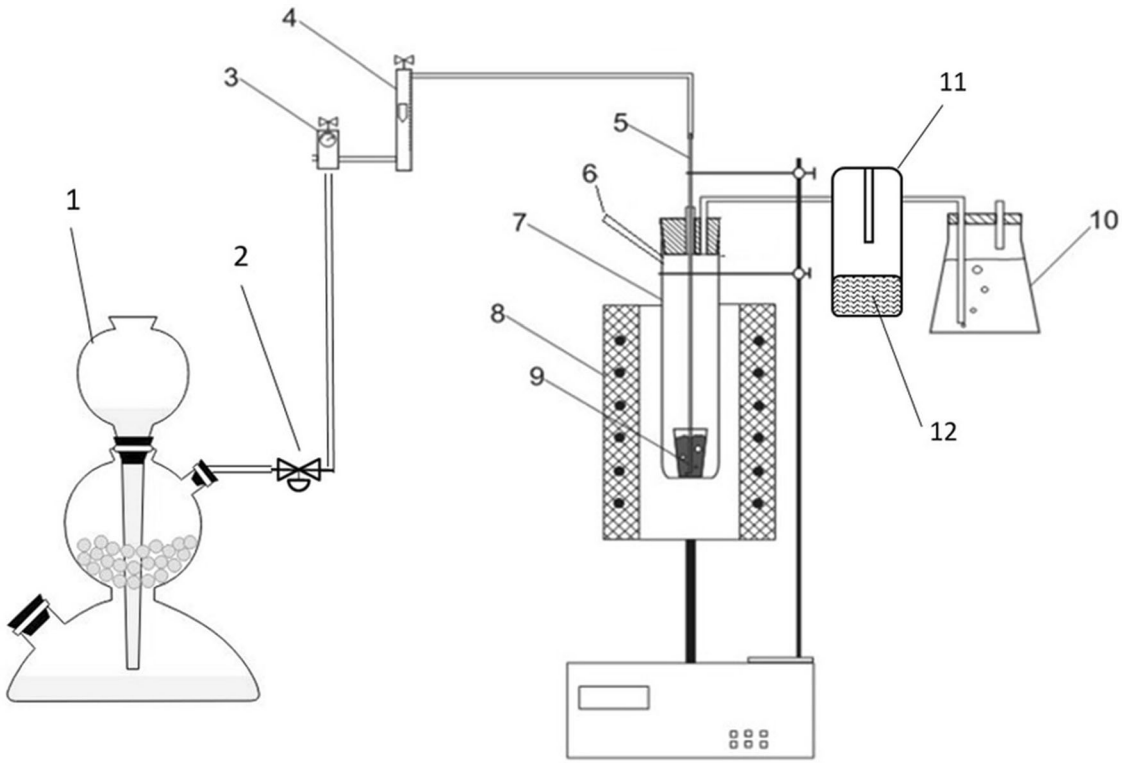



Fig. 2. Laboratory reactor: (1) chlorine generator; (2) gas stop; (3) valve; (4) flowmeter; (5) tube for chlorine; (6) place for air addition; (7) quartz reactor; (8) furnace with temperature controller; (9) crucible; (10) gas cleaning bottle, (11) sublimate collector, (12) sublimates.

strong absorption of the RE metals and iron at 8 KeV. Although Pr is present in both magnets (as EDS and ICP-MS will show in the following section), since the XRD databases do not contain profiles for amounts of Pr intermediate between none and equimolar Nd/Pr, we present the ICSD standard profile for $Nd_2Fe_{14}B$ as the better comparison, particularly in the low angle region. We would also point out, however, that standard XRD profiles are obtained from crystalline powders with randomly oriented grains, whereas that is certainly not the case for the RE-Fe-B magnets. As a result, unlike XRD peak positions, peak intensities cannot be required to match.

Each magnet was placed in the scanning electron microscope on carbon support and the energy dispersive spectra (EDS) measured in reflection. Similar to the case of the XRD measurements, only a thin layer (1–2 $\mu$m) at the fragment surface is probed. We note that magnet#1 does indeed contain Dy dopant, in addition to Pr, while no Dy can be detected for magnet#2; rather Ce is present. We had expected to find Co accompanying Ce, as suggested in Ref. 3; however no x-ray fluorescence (XRF) peak for Co was detected in the EDS spectrum. There was however a weak peak characteristic of Ca, for which we currently have no explanation. An aluminum protective coating appears to have been used for both magnets;[20] however, prior to shipment from Korea, the magnet surface had been polished, which may explain the extensive corrosion observed in the photograph of magnet #1. In addition, given that $AlCl_3$ boils at 453 K, some of the Al may have oxidized or formed hydroxides during the preparation or reaction, which may explain why an Al signal remains in the clinker samples. We should mention that the small amount of Al detected could also be intrinsic to the sample (see Ref. 20). ICP-MS elemental analysis (mass%) of 30 g as-received magnet #1 gives: Nd, 23.5; Pr,7.8; Dy, 3.7: Ce, 0.01; Fe, 61.7; A, 0.6. For 28 g as-received magnet #2, the corresponding amounts are: Nd, 20.3; Pr, 8.5; Dy,0.2; Ce,8.5; Fe,51.5; Al, 8.9 (Fig. 3).

## Thermodynamic Calculations for Chlorine Gas Reacting with Nd Magnets

Calculations of Gibbs energy were performed using software developed by some of the authors and based on standard values for pure substances.[25] The Gibbs energy ($\Delta G$) values for chlorination reactions within the temperature range 473–773 K are shown in Table I.

All elements likely to be present in the magnet fragments (iron, neodymium, praseodymium, dysprosium, cerium, boron, aluminum) may be expected to be chlorinated in accordance with reactions (1–8) in Table I upon injection of chlorine gas. Chlorides of iron and boron, and also of aluminum, are relatively volatile (boiling point of $FeCl_3$, 589 K; of $BCl_3$, 284 K; of $AlCl_3$, 453 K) while

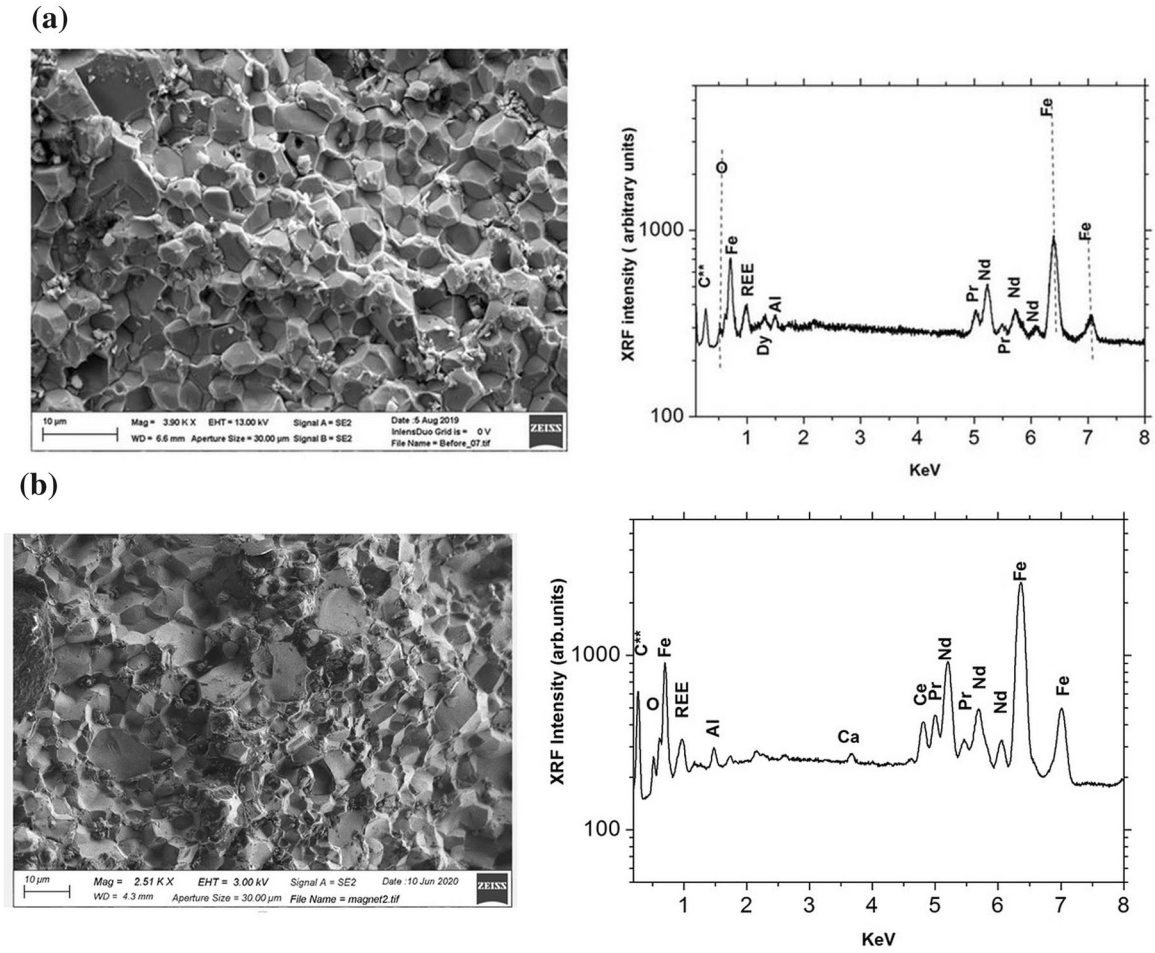



Fig. 3. SEM micrographs and EDS fluorescence spectra from the surface (1–2 $\mu$m penetration depth) of: (a) as-received magnet # 1; (b) as-received magnet #2.

**Table I. Gibbs energy ($\Delta G$) calculated for reaction of metals and aluminum oxide during moderate to high temperature treatment with $Cl_2$ gas**

| | | Temperature, K | | | |
|---|---|---|---|---|---|
| | | 473 | 573 | 673 | 773 |
| N | Reaction | $\Delta G$, kJ/mole | | | |
| 1 | Fe (s) + 1.5 $Cl_2$ (g) = $FeCl_3$ (s) | − 295.8 | − 275.9 | − 263.0 | − 251.3 |
| 2 | Fe (s) + 1.5 $Cl_2$ (g) = $FeCl_3$ (g) | − 242.3 | − 240.5 | − 238.6 | − 236.6 |
| 3 | Nd (s) + 1.5 $Cl_2$ (g) = $NdCl_3$ (s) | − 923.4 | − 899.4 | − 875.8 | − 852.7 |
| 4 | B (s) + 1.5 $Cl_2$ (g) = $BCl_3$ (g) | − 379.1 | − 374.1 | − 369.0 | − 364.0 |
| 5 | Pr (s) + 1.5 $Cl_2$ (g) = $PrCl_3$ (s) | − 937.1 | − 913.0 | − 889.3 | − 866.0 |
| 6 | Dy(s) + 1.5 $Cl_2$ (g) = $DyCl_3$ (s) | − 863.3 | − 836.3 | − 809.6 | − 783.2 |
| 7 | Ce (s) + 1.5 Cl2 (g) = $CeCl_3$ (s) | − 934.5 | − 910.4 | − 886.7 | − 863.5 |
| 8 | Al (s)+1.5$Cl_2$ (g) = $AlCl_3$ (g) | − 561.4 | − 556.4 | − 551.3 | − 546.2 |
| 9 | $Al_2O_3$(s) + 3$Cl_2$(g) = 2$AlCl_3$(g) + 1.5$O_2$(g) | 96.6 | 91.5 | 86.4 | 81.4 |

the rare earth metal chlorides remain in the clinker because of their extremely high boiling points (boiling point of $NdCl_3$, 1873 K; of $PrCl_3$, 1983 K; of $DyCl_3$, 1803 K; of $CeCl_3$, 2000 K). Air flowed across the upper part of the reactor for ferric chloride oxidation is in accordance with reaction (2) or (3) detailed below:

$$2\ FeCl3 + 1.5O_2 = Fe_2O_3 + 3Cl_2 \qquad (2)$$

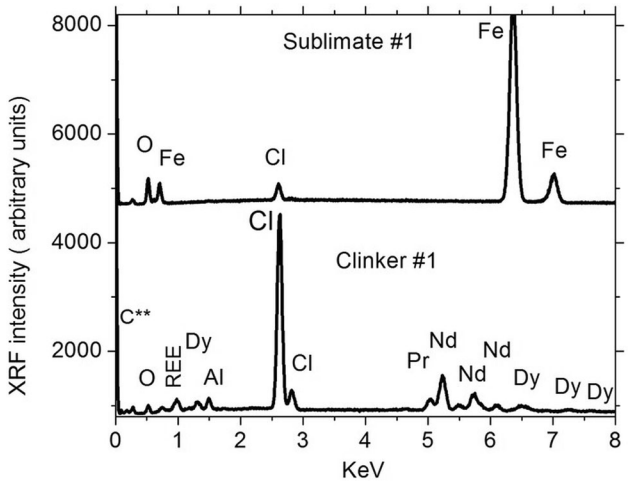



Fig. 4. The clinker XRF spectrum compared with that of the sublimate material following treatment of magnet #1for 2 h at 673 K in flowing Cl gas.

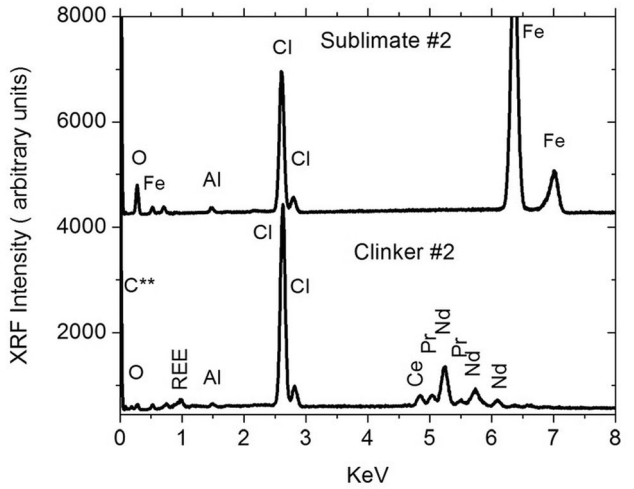



Fig. 5. The clinker XRF spectrum compared with that of the sublimate material following treatment of magnet #2 for 2 h at 673 K in flowing Cl gas.

$$FeCl3 + 0.5O_2 = FeOCl + Cl_2 \qquad (3)$$

Reaction (2) is marginally more thermodynamically favorable near room temperature: $\Delta G$ (reaction 2) $\approx -40$ kJ/mole and $\Delta G$ (reaction 3) $\approx -30$ kJ/mole. Chlorine gas produced by either reaction (2) or (3) can be returned in the pilot or industrial apparatus to the chlorination unit; therefore, with proper design, recycling of most of the chlorine gas may be achieved. After cooling under nitrogen flow, the crucible was removed from the furnace and broken. The final products—clinker powders—obtained from magnets #1 and #2 appear off-white to pale yellow in fluorescent light. Both were immediately weighed (18 g and 17 g, respectively) and analyzed with powder x-ray diffraction (XRD) (Fig. 6) as well as energy-dispersive spectroscopy (EDS) in the scanning electron microscope (SEM) (Figs. 4, 5). Dark brown powder was also collected from the top part of the reactor and similarly analyzed with EDS (Figs. 4 and 5) and XRD (Fig. 7a and b).

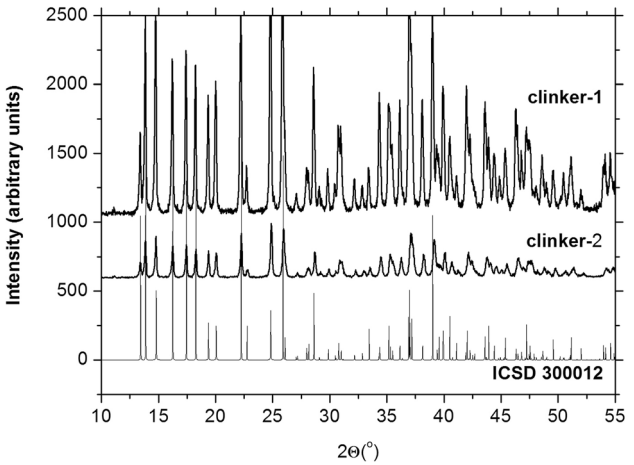



Fig. 6. XRD patterns of the crystalline clinker powders obtained following chlorine gas treatment of the magnet fragments (#1, #2) at 673 K. The ICSD database diffraction pattern for $NdCl_3$ hexahydrate is presented for identification.

## Clinker Elemental Composition and Crystal Structure

The elemental composition of the clinker powders which remain in the crucible following treatment of the Nd magnets with chlorine gas for 2 h at 673 K was characterized by energy dispersive x-ray fluorescence spectroscopy (Figs. 4 and 5) and by ICP-MS following dissolution in aqua regia. In Figs. 4 and 5, the strong doublet due to chlorine dominates the spectrum while in neither case was the x-ray fluorescence peak of Fe detected. However, the set of REEs that were present in each of the as-received samples was also present in the resulting clinker powders. ICP-MS quantitation of the elemental composition (mass%) for 18 g gives clinker #1: Nd, 39.5; Pr, 13.2; Dy,6.0: Ce, 0.01; Al 0.06. For 17 g clinker #2, the corresponding amounts are: Nd, 32.3; Pr, 13.0; Dy, 0.3; Ce, 13.5; Al, 0.1. In neither sample was any significant amount of Fe detected by ICP-MS. On the basis of these data, the material balance for chlorine gas treatment of Nd magnets was calculated (see Table ESM1, Table ESM2). Approximate extraction yield of rare earth metals to the clinkers was: Nd: 99%; Pr: 95–99%; Dy: 93–96%; Ce: 92–96%. Iron was almost completely removed to sublimates; the iron extraction yield to clinker was 0.01–0.04%.

The x-ray diffraction patterns of the clinker powders identify them as being dominated by $NdCl_3$ hexahydrate (ICSD #300012) (Fig. 6). It is known that $NdCl_3$ is highly hydroscopic and readily absorbs water when exposed to the atmosphere at ambient temperature. The similar chemistry of the

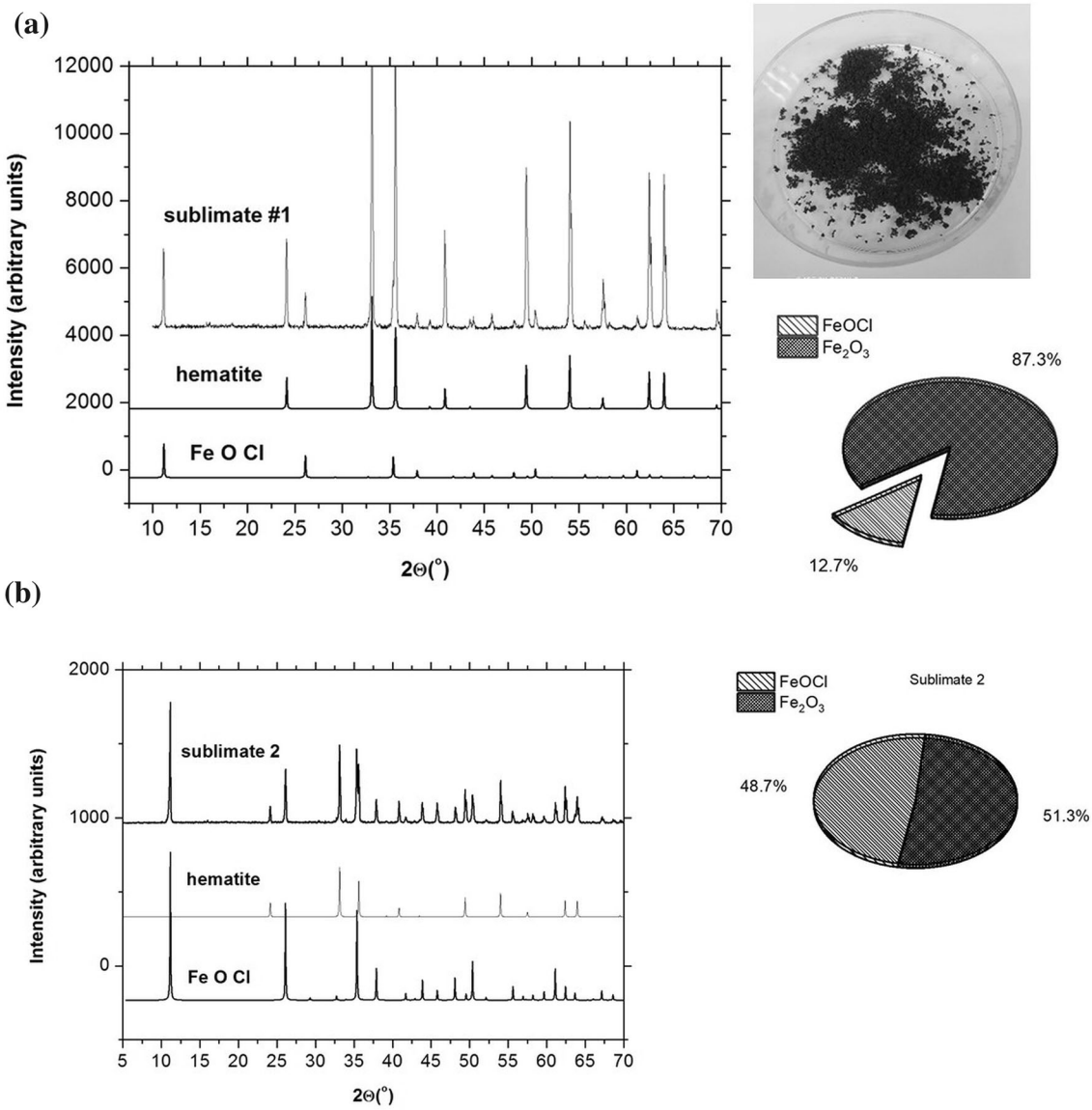



Fig. 7. (a) Powder x-ray diffraction (XRD) patterns, color photograph and quantitative phase analysis of the volatile material from Nd magnet sample #1 following 2 h of treatment of the magnet fragments at 673 K with chlorine gas. (b) Powder x-ray diffraction (XRD) patterns and quantitative phase analysis of the volatile material from Nd magnet sample #2 following 2 h of treatment of the magnet fragments at 673 K with chlorine gas. The sublimates from magnets #1 and #2 are visually similar.

trichlorides of the RE dopants—Pr, Dy and Ce—results in crystals with the same symmetry (P2/n) and very similar lattice spacings (monoclinic unit cell: $a$ = 7.9–8.0Å; $b$ = 6.6–7.9Å; $c$ = 9.6–9.7Å, $\beta$ = 93.5–93.7°) and hygroscopic tendency as the host Nd trichloride. Whole pattern fitting (of the first 13 peaks) of the clinker1 powder to the literature $NdCl_3$ hexahydrate diffraction pattern gave an R factor of $\sim$ 12% and could not distinguish any other crystalline phases (see Table ESM3 and Fig. ESM1). In view of the fact that the melting points of the various RE chlorides are so similar (melting point of $NdCl_3$ is 1026 K; of $PrCl_3$, 1059 K; of $DyCl_3$, 920 K; of $CeCl_3$, 1090 K), no attempt was made to effect preferential separation of the rare earth metals via molten salt extraction. Additionally, in the interests of energy economy, we preferred to avoid such high temperatures.

Moving to the characterization of the composition of the volatile material that accumulated at the top of the reactor, predominantly three elements contribute to the EDS spectra: iron, chlorine and oxygen (Figs. 4 and 5). Given the very low boiling point of boron chloride (284 K), any boron chloride that might have been present initially would have evaporated prior to the EDS or XRD measurements. $BCl_3$ is an aggressive toxic reagent. It is fully hydrolyzed with water, releasing heat and producing solid $B(OH)_3$ according to $BCl_3 + 3H_2O = B(OH)_3 + 3HCl$. XRD finds two crystal structures for the sublimate powders—hematite $Fe_2O_3$ (ICSD#15840) and iron (III) oxychloride (FeOCl) (ICSD#16013)

(Fig. 7a and b). Making use of the reference intensity ratio (RIR) parameters available in JADE_PRO for these two structures, quantitative phase analysis of the x-ray diffraction patterns reveals that hematite dominates in the sublimate powder from magnet #1, while in the case of magnet #2, hematite and iron oxychloride are present in almost equal amounts. We believe that the rate of air flow at the top of the reactor is the controlling factor for the relative amount of oxychloride present. As might have been expected from the low boiling point of $AlCl_3$ (453 K), small amounts of Al are found in the sublimate (as well as in the clinker), and this is true for both magnets. Thermodynamic calculations (not shown) demonstrate that nickel and copper, present as anti-corrosion coatings, would also be chlorinated at the reaction temperatures used here and transformed into metal chlorides; however, Ni and Cu chlorides, being solids, could indeed contaminate the clinker.

### Reducing RE Chlorides to RE Metals

Since RE metal chlorides must be reduced so that the rare earth metals may be usefully recycled, we suggest two procedures which can accomplish that task, both of which are known to perform well on an industrial scale: (1) electrolysis of the molten salts for rare earth metal production.[26,27] It was found that on an inert electrode (W or Mo), $Nd^{3+}$ ions are reduced to metallic neodymium through two consecutive steps:

$$Nd^{3+} + e- \rightarrow Nd^{2+} \quad (4a)$$

followed by

$$Nd^{2+} + 2e- \rightarrow Nd \quad (4b)$$

A second option (2) is metallothermic reduction accomplished by mixing the rare earth trichloride with alkali or alkaline earth metals in vacuum or argon atmosphere at 573–673 K.[28–30] With excess Li, for example, this results in rare earth metal and LiCl:

$$NdCl_3 + 3Li \rightarrow Nd + 3LiCl \quad (\Delta G \approx -180\ \text{kJ/mole}) \quad (5)$$

The electrolytic pathway is generally considered to be simpler to perform and less costly than metallothermic reduction procedures. However, detailed investigation and comparison of these procedures is beyond the scope of the present study.

## CONCLUSIONS

We have described a laboratory-scale process for recycling rare earth metals from end-of-life Nd/Pr-iron-boron sintered-alloy permanent magnets using a chlorine gas treatment during 2 h at 673 K. This treatment does not require any special preparation of the magnets; they can be used without demagnetization, crushing or milling, although the presence of an intact protective coating may require protocol modification. Following treatment at 673 K, a clinker powder consisting of hydrated rare earth metal chlorides is obtained, while hematite and iron oxychloride sublimate. Referenced literature confirms that the rare earth metal chlorides in the clinker may be readily reduced to metal by electrolysis in relatively low temperature eutectic melts or, alternatively, by metallothermic reduction in mixtures with, for example, excess lithium.

## FUNDING

The authors have no financial relationship with the organization that sponsored the research.

## CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

## SUPPLEMENTARY INFORMATION

The online version contains supplementary material available at https://doi.org/10.1007/s11837-021-04592-3.

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